Integration

Workshop – Term 3, Week 7

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Outline

1 Housekeeping

2 Floor and Ceiling Functions

3 Symmetry

4 Feynman Technique

5 Gamma and Beta Functions

6 Appendix and resources

Attendance!

Please scan the attendance form so you can get free food



Assumptions

I will assume you at least know basic integration techniques from extension 2 / math 1b, including mainly:

1. U-substitution
2. Trig substitution
3. Integration by parts
4. Basic partial fractions
5. Integrals of elementary functions (polynomials, trig, hyperbolic trig, etc.)

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DISCLAIMER!!

This workshop is NOT a fully comprehensive guide to integration! Most integration problems are solved on a case-by-case basis; this workshop just covers a few common / basic cases.

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Floor and Ceiling Functions

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Let x be a real number. Then the floor function of x, denoted $\lfloor x \rfloor$, is the largest integer less than or equal to x, and the ceiling function of x, denoted $\lceil x \rceil$ is the smallest integer greater than or equal to x.

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Example

$$\lfloor \pi \rfloor = 3$$
$$\lceil 67 \rceil = 67$$
$$\lceil 1.00000000000000000001 \rceil = 2$$

Floor and Ceiling Functions

In general, whenever you spot a floor or ceiling function, you should split up the integral into intervals between consecutive integers; this is because the floor and ceiling functions are constant between any two consecutive integers.

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Example

$$\int_0^{2027} \sin\left(\frac{\pi}{2}\lfloor x \rfloor\right) dx$$

Floor and Ceiling Functions

Solution

$$\begin{aligned}\int_0^{2027} \sin\left(\frac{\pi}{2}\lfloor x\rfloor\right) dx &= \sum_{n=0}^{2026} \int_n^{n+1} \sin\left(\frac{\pi}{2}\lfloor x\rfloor\right) dx \\ &= \sum_{n=0}^{2026} \int_n^{n+1} \sin\left(\frac{\pi n}{2}\right) dx = \sum_{n=0}^{2026} \sin\left(\frac{\pi n}{2}\right) \\ &= \sum_{k=0}^{1013} \sin(\pi k) + \sum_{k=0}^{1012} \sin\left(\frac{(2k+1)\pi}{2}\right) \\ &= 0 + \sum_{k=0}^{1012} (-1)^k = 1\end{aligned}$$

Example on Floor Functions

Example

$$\int_0^\infty e^{-\lfloor x \rfloor} dx$$

Example on Floor Functions

Solution

$$\begin{aligned}\int_0^\infty e^{-\lfloor x \rfloor}\,dx &= \sum_{n=0}^{\infty}\int_n^{n+1} e^{-\lfloor x \rfloor}\,dx \\ &= \sum_{n=0}^{\infty}\int_n^{n+1} e^{-n}\,dx \\ &= \sum_{n=0}^{\infty} e^{-n} = \sum_{n=0}^{\infty}(e^{-1})^n \\ &= \frac{1}{1-\frac{1}{e}} = \frac{e}{e-1}\end{aligned}$$

In General...

In general, an integral with a floor / ceiling function will reduce to some sort of sum (whether finite or infinite). It would be handy to know some common summations, mainly

1. Arithmetic progressions
2. Geometric Series (finite and infinite)
3. Sum of first n integers, squares and cubes
4. $\zeta(2) = \sum_{n=1}^{\infty} 1/n^2 = \pi^2/6$

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King's Rule

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For f continuous on $[a, b]$,

$$\int_a^b f(x)\,dx = \int_a^b f(a+b-x)\,dx.$$

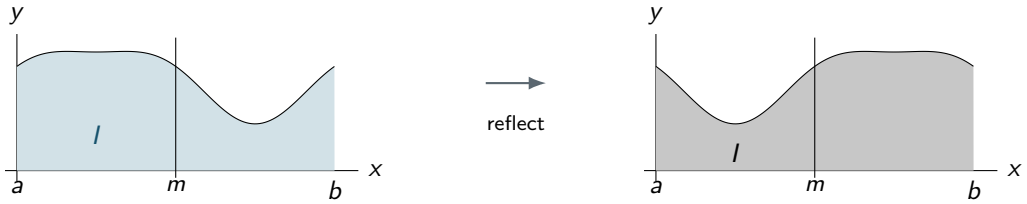
King's Rule

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For f continuous on $[a, b]$,

$$\int_a^b f(x)\,dx = \int_a^b f(a+b-x)\,dx.$$

To understand why, substitute $u = a + b - x$. As x runs from a to b, u runs from b to a, the minus sign in $du = -dx$ compensates for the reversed endpoints.



Reflection about the midpoint preserves the area.

Queen's Rule

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For f continuous on $[0, 2a]$,

$$\int_0^{2a} f(x)\,dx = \int_0^{a} \big[f(x) + f(2a - x)\big]\,dx.$$

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To understand why, split the integral at a:

$$\int_0^{2a} f(x)\,dx = \int_0^a f(x)\,dx + \int_a^{2a} f(x)\,dx.$$

In the second integral, we substitute $x = 2a - u$:

$$\int_a^{2a} f(x)\,dx = \int_0^a f(2a - u)\,du.$$

Renaming the variable u as x, we obtain Queen's rule.

Integrals of Periodic Functions

We recall that f is T-periodic if $f(x+T) = f(x)$ for all x.

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Result

For f continuous and T-periodic, and for every $a \in \mathbb{R}$ and $n \in \mathbb{N}$,

$$\int_a^{a+T} f(x)\,dx = \int_0^T f(x)\,dx, \qquad \int_0^{nT} f(x)\,dx = n\int_0^T f(x)\,dx.$$

We can read this as saying that every interval of length T contains exactly one full period, no matter where it begins. If we split $[0, nT]$ into n consecutive periods, each contributes the same integral, so we obtain the second identity.

Integrals of Periodic Functions

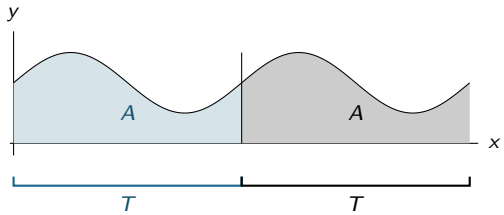
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The Standard π-Integral

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The Standard π-Integral

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Solution

We set $f(x) = \frac{\sin x}{1+\cos^2 x}$ *and* $P = \int_0^{\pi} xf(x)\, dx$. *Applying King's rule on* $[0, \pi]$, *we find*

$$I = \int_0^{\pi} (\pi - x) f(x)\, dx = \pi \int_0^{\pi} f(x)\, dx - I,$$

so $I = \frac{\pi}{2} \int_0^{\pi} f(x)\, dx$.

The Standard π-Integral

Solution

Because $f(\pi - x) = f(x)$, Queen's rule halves the interval, with $u = \cos x$, we obtain

$$\int_0^{\pi} f(x)\,dx = 2\int_0^{\pi/2} f(x)\,dx = 2\int_0^{1} \frac{du}{1+u^2} = \frac{\pi}{2}.$$

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Therefore

$$\int_0^{\pi} f(x)\,dx = \frac{\pi}{2}, \qquad I = \frac{\pi^2}{4}.$$

The 2026π Integral

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The 2026π Integral

Solution

We can check that f is 2π-periodic and that $\int_0^{2\pi} f(x)dx = 0$, so we can split the integral into 2π long intervals, that is

$$\begin{aligned} I &= \int_0^{2\pi} xf(x)dx + \int_{2\pi}^{4\pi} xf(x)dx + \cdots + \int_{2024\pi}^{2026\pi} xf(x)dx \\ &= \sum_{k=0}^{1012} \int_{2k\pi}^{2(k+1)\pi} xf(x)dx. \end{aligned}$$

The 2026π Integral

Solution

If we make the substitution $x = 2k\pi + u$, then we get that

$$\begin{aligned} I &= \sum_{k=0}^{1012} \int_0^{2\pi} (u + 2k\pi) f(u) du \\ &= \sum_{k=0}^{1012} \int_0^{2\pi} u f(u) du + \sum_{k=0}^{1012} \int_0^{2\pi} 2k\pi f(u) du \\ &= 1013 \int_0^{2\pi} u f(u) du. \end{aligned}$$

The 2026π Integral

Solution

Now we can use Queen's rule on the new integral, noting that $f(2\pi - u) = -f(u)$ and the previous integrals we did, we get that

$$\begin{aligned}\int_0^{2\pi} uf(u)du &= \int_0^{\pi} uf(u) + (2\pi - u)f(2\pi - u)du \\ &= \int_0^{\pi} 2uf(u) - 2\pi f(u)du \\ &= 2 \cdot \frac{\pi^2}{4} - 2\pi \cdot \frac{\pi}{2} = -\frac{\pi^2}{2},\end{aligned}$$

and so $I = -\frac{1013\pi^2}{2}$.

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Leibniz's Rule

We want a way to differentiate a variable under the integral sign. Recall that $\frac{\partial f}{\partial t}(x, t)$ is the derivative of f with respect to t, holding x fixed.

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Let $f : [a,b] \times [c,d] \to \mathbb{R}$ be continuously differentiable. Then

$$\frac{d}{dt}\int_a^b f(x,t)\,dx = \int_a^b \frac{\partial f}{\partial t}(x,t)\,dx.$$

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$$\frac{d}{dt}\int_a^b f(x,t)\,dx = \int_a^b \frac{\partial f}{\partial t}(x,t)\,dx.$$

The idea is to introduce a parameter t in the integral and call it $I(t)$. We can then use the rule above to find $I'(t)$ and solve a differential equation.

Note that this technique of using Leibniz's Rule is often called the Feynman method.

Leibniz Example 1

The first step would be to identify how to introduce the parameter in a way that works out.

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In this case, it is a good idea to let $I(t) = \int_0^1 \frac{x^t - 1}{\ln x}\, dx$. We want $I(3)$.

Leibniz Example 1

Solution

Using leibniz' rule, we obtain

$$\begin{aligned} I'(t) &= \int_0^1 \frac{\partial}{\partial t}\left(\frac{x^t - 1}{\ln x}\right) dx \\ &= \int_0^1 \frac{x^t \ln x}{\ln x}\, dx \\ &= \frac{1}{t+1}. \end{aligned}$$

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So then $I(t) = \ln|t+1| + C$, *for a constant* C. *Clearly,* $I(0) = 0$, *so* $C = 0$. *We thus have* $I(3) = \ln 4$.

Leibniz Example 2

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Harder Example

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One observation to make is that the problematic part of this integral is the numerator - $\frac{1}{1+x^2}$ is easy to integrate, whilst $\frac{1}{1+x^t}$ is not.

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One observation to make is that the problematic part of this integral is the numerator - $\frac{1}{1+x^2}$ is easy to integrate, whilst $\frac{1}{1+x^t}$ is not. A suitable choice of parameter is

$$I(t) = \int_0^\infty \frac{\tan^{-1}(tx)}{x(1+x^2)}\, dx.$$

Some reassurance that this may work is that $I(0)$ is easy to calculate in order for us to work out the constant.

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Using leibniz' rule, we obtain

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Solution

Using leibniz' rule, we obtain

$$\begin{aligned} I'(t) &= \int_0^\infty \frac{\partial}{\partial t}\left(\frac{\tan^{-1}(tx)}{x(1+x^2)}\right) dx \\ &= \int_0^\infty \frac{1}{x(1+x^2)} \cdot \frac{x}{1+t^2x^2}\, dx \end{aligned}$$

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Leibniz's Rule Example 2: Tangent

To evaluate this integral, we will need the following result.

Result

For any $a > 0$, we have

$$\int_0^\infty \frac{a}{1 + a^2x^2}\, dx = \frac{\pi}{2}.$$

Solution

Let $ax = \tan\theta$. *Then*

$$\int_0^\infty \frac{a}{1 + a^2x^2}\, dx = \int_0^{\frac{\pi}{2}} \frac{\sec^2\theta}{1 + \tan^2\theta}\, d\theta = \int_0^{\frac{\pi}{2}} d\theta = \frac{\pi}{2}.$$

leibniz Rule Example 2

Solution

Evaluating the integral we get

$$\frac{1}{t^2-1}\left(\frac{\pi}{2}+\frac{t\pi}{2}\right)=\frac{\pi}{2(1+t)}.$$

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From here we get $I(t)=\frac{\pi}{2}\ln|1+t|$ *and thus* $I=\frac{\pi}{2}\ln 2$.

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Solution

Evaluating the integral we get

$$\frac{1}{t^2-1}\left(\frac{\pi}{2}+\frac{t\pi}{2}\right)=\frac{\pi}{2(1+t)}.$$

From here we get $I(t)=\frac{\pi}{2}\ln|1+t|$ *and thus* $I=\frac{\pi}{2}\ln 2$.

Some warning signs of when this is practical to think about on an integral are when you have complicated trigonometric/exponential functions on the numerator and a polynomial denominator - the chain rule may allow an x term to cancel out.

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Gamma and Beta Functions

Definition

Define the gamma and beta functions by

$$\Gamma(z) = \int_0^\infty t^{z-1} e^{-t} dt$$

$$B(m,n) = \int_0^1 x^{m-1}(1-x)^{n-1} dx, m, n > 0$$

Gamma and Beta Functions

Properties of the Gamma Function

1. $\Gamma(n+1) = n!, n \in \mathbb{N}$
2. $\Gamma(1/2) = \sqrt{\pi}$
3. $\Gamma(n+1) = n\Gamma(n)$

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Properties of the Beta Function

1. $B(m,n) = B(n,m)$
2. $B(m,n) = \frac{\Gamma(m)\Gamma(n)}{\Gamma(m+n)}$
3. $B(m+1,n) = \frac{m}{m+n}B(m,n)$

$$B(m,n) = \int_0^\infty \frac{x^{m-1}}{(1+x)^{m+n}}dx$$

Example on Gamma Functions

Example

$$\int_0^1 x^2(\ln(x))^3 dx$$

Example on Gamma Functions

Solution

We make the substitutions $x = e^{-u}$ *and* $t = 3u$ *such that*

$$\begin{aligned}\int_0^1 x^2(\ln(x))^3 dx &= \int_\infty^0 e^{-2u}(-u)^3(-e^{-u})du \\ &= -\int_0^\infty u^3 e^{-3u} du \\ &= -\frac{1}{3^4}\int_0^\infty t^3 e^{-t} dt \\ &= -\frac{\Gamma(4)}{3^4} = \frac{-3!}{3^4} = \frac{-2}{27}\end{aligned}$$

Example on Beta Functions

Example

$$\int_0^{\frac{\pi}{2}} \sin^3(x)\cos^7(x)dx$$

Example on Beta Functions

Solution

We make the substitution $u = \sin^2(x)$ *such that*

$$\begin{aligned}\int_0^{\frac{\pi}{2}} \sin^3(x)\cos^7(x)dx &= \frac{1}{2}\int_0^1 u(1-u)^3 dx \\ &= \frac{1}{2}B(2,4) \\ &= \frac{1}{2}\frac{\Gamma(2)\Gamma(4)}{\Gamma(6)} = \frac{1!3!}{2\cdot 5!} \\ &= \frac{1}{40}\end{aligned}$$

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Feedback!

Please fill out feedback form so we know how to run future workshops :D



(a) Feedback form



(b) Attendance form

Photo time :)